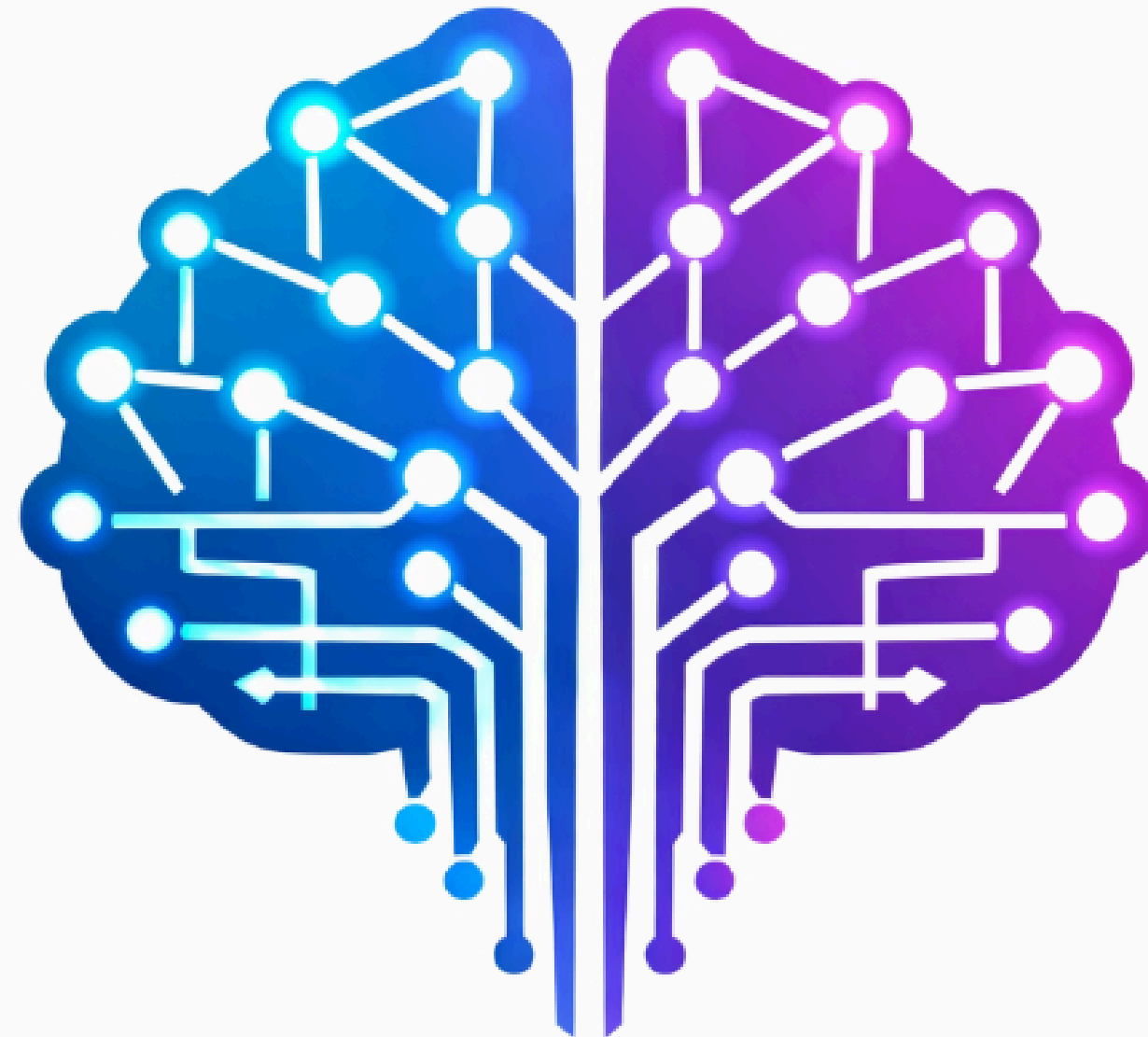


GENAI FUNDAMENTALS



LECTURE:
PROMPTING FUNDAMENTALS

What Is a Prompt?

A prompt is the input given to an LLM to guide its response.

A prompt may include:

- instructions
- context
- examples
- formatting requirements

Better prompts usually produce **better outputs**.

Why Prompt Design Matters

Small changes in prompts can significantly affect outputs.

Example Prompt-1: "Explain machine learning."

Example Prompt-2: "Explain machine learning to a beginner using simple analogies."

The second prompt produces a **more useful response**.

The Basic Structure of a Good Prompt

Effective prompts often contain three components:

- **Instruction:** What the model should do.
- **Context:** Relevant information for the task.
- **Output format:** How the response should be structured.

Few-Shot Prompting

LLMs can learn tasks from examples in the prompt.

Example:

Input → Output examples guide the model.

This technique is called **few-shot prompting**.

Asking for Structured Outputs

Prompts can instruct models to produce **structured responses**.

Examples:

- bullet points
- tables
- JSON format
- step-by-step explanations

Structured outputs make results **easier to process programmatically**.

Iterative Prompting

Prompting is often an **iterative process**.

Developers frequently refine prompts by:

- adjusting instructions
- adding context
- changing output format
- providing examples

Prompt iteration improves reliability.

Key Takeaway

Prompting helps developers guide LLM behavior.

Key principles:

- provide clear instructions
- include relevant context
- define output structure
- use examples when needed
- refine prompts through iteration

Prompt design is a **core skill for GenAI developers**.

